

EXECUTIVE AND TECHNICAL REPORT

The store is losing revenue on its product pages and after the holidays, and its tracking hides which channels really sell

An analysis of the GA4 public sample for the Google Merchandise Store, 1 November 2020 to 31 January 2021, with the historical data corrected for four tracking faults.



The question. Where is the Google Merchandise Store losing revenue, which channels really drive sales, and what should it fix or test first?



\$338.3K

Revenue, deduplicated



5,288

Orders



360.1K

Sessions



1.47%

Conversion rate
(orders per session)

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PART 1



Executive summary

Five findings, each with how far the data supports it and what to do about it. Confidence is **High** when measured on all the data and **Medium** when it rests on a short or seasonal window.



The raw tracking would have misled every report. Four faults, including 30% of orders credited to the store's own website.

High: measured on all the data

→ Fix GA4 at source and alert when a tag stops firing



3 in 4 product viewers never add to basket, with no clear difference by device. Four big entry pages bring 29% of visits but 7% of purchases.

High for the gap; cause not yet known

→ Check those pages' traffic, then A/B test the product page (about two weeks)



Organic Search is credited with 40 to 43% of sales under all seven attribution models.

High for credit; credit is not cause

→ Protect SEO; don't cut Paid Search on last-click numbers



January revenue per day fell 57% while sessions fell 11%: conversion weakened.

Medium: one holiday season

→ Plan Q1 on a post-holiday baseline



The store runs on one-off purchases. 7.5% of customers buy on more than one day; repeat buyers return within about two weeks.

Medium: three months, browser-level IDs

→ Test a second-purchase offer against a holdout (about three months)



What happens next

The analysis can locate the problems but not prove the fixes, so every recommendation comes with a test sized from the store's own traffic (Part 3). The first two are the tracking fix, which can be verified within a week, and a product-page A/B test, which needs about two weeks to detect a 10% lift.

PART 2 · FINDINGS



1. Measurement reliability

30%

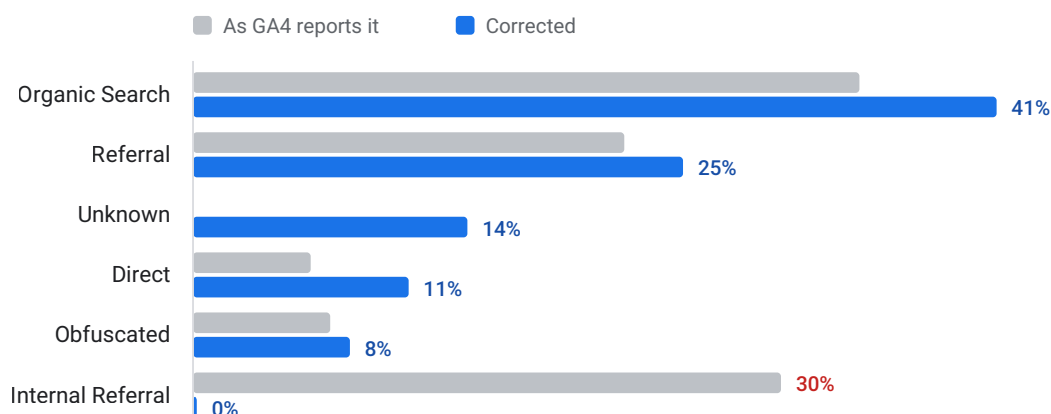
of orders were credited to the store's own website. Total revenue moves only 7% once the tracking is fixed, but the channel and funnel picture changes completely.

FAULT	DISTORTS	EFFECT IF IGNORED	HOW IT WAS DETECTED	TREATMENT
Self-referrals	Channel credit	30% of orders credited to the store itself; Organic Search understated by 19%	1,589 orders credited to the store's own domain	Credit goes to the visits before; 719 with no earlier visit are Unknown
Missing transaction IDs	Revenue and orders	\$30.7K of real orders (837) dropped by the usual dedupe	906 purchase events with no ID, mostly early November	Same session, amount and basket = one order
Duplicate purchases	Revenue	+7% revenue (\$23.8K)	404 of 5,692 events repeat an order	One event per order
Add-to-cart tag broken	Funnel	View to cart 19.7% instead of 25.7%; cart to checkout 73% instead of 53%	22 November days with almost no cart events while checkouts continued	Funnel uses the other 70 days
Order values missing	January revenue	January looks -64% per day instead of -57%	26 to 31 January: values collapse to \$0	Value comparisons skip those days

Only the 906 purchases without an ID are ambiguous: depending on how they're treated, orders fall between 4,451 and 5,357 (appendix B).

Self-referrals: 870 orders go back to real channels, 719 have no known source

Share of last-click orders by channel, before and after the corrections



Share of last-click orders by channel, as GA4 reports them (grey) and corrected (blue). Observed. Source: GA4 public sample, 1 Nov 2020 to 31 Jan 2021.

FINDING

Four faults distort channel and funnel reporting; total revenue barely moves.

EVIDENCE

30% of orders miscredited; view to cart 6 points too low on broken days.

INTERPRETATION

Reports on the raw data would point the team at the wrong channel and the wrong funnel step.

→ NEXT ACTION

Unwanted referrals, an ID on every purchase, alerts on tag volume (Part 3, test 1).



2. Which channels get credit for sales

40–43%

Organic Search is credited with 40 to 43% of sales under all seven attribution models, so the result doesn't depend on the method.

Organic Search gets 40 to 43% of conversions under every model

Share of conversions credited to each channel, by attribution model · darker means more credit

	Last click	Last non-direct	First click	Linear	Time decay	Position-based	Data-driven
Organic Search	40.9%	43.0%	42.6%	41.4%	41.6%	41.6%	40.0%
Referral	24.9%	25.8%	24.2%	24.2%	24.6%	24.4%	23.3%
Direct	10.7%	6.8%	11.1%	11.4%	11.0%	11.1%	13.2%
Unknown	13.6%	13.6%	13.6%	13.6%	13.6%	13.6%	12.2%
Obfuscated	8.2%	9.0%	6.6%	7.6%	7.5%	7.5%	9.2%
Paid Search	1.2%	1.3%	1.5%	1.3%	1.3%	1.3%	1.5%
Email	0.5%	0.5%	0.4%	0.5%	0.5%	0.5%	0.5%
Affiliates	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.2%

Share of order credit by channel and model. Modelled. Position-based uses a 40/20/40 split; data-driven is a Markov chain.

DATA-DRIVEN (MARKOV) CREDIT	SHARE	95% INTERVAL	VS LAST CLICK
Organic Search	40.0%	38.6% to 41.1%	-2%
Referral	23.3%	22.4% to 24.3%	-6%
Direct	13.2%	12.6% to 13.8%	+23%
Unknown (returned from payment, no earlier visit)	12.2%	11.3% to 13.0%	-11%
Paid Search	1.5%	1.2% to 1.8%	+23%

65% of orders come from a single visit, where every model agrees, so the model only matters for about a third of orders. Paid Search gets 23% more credit from the data-driven model than from last click (79 against 64

orders): it tends to start journeys others finish. That's a reason not to cut it on last-click numbers, not proof that it causes sales, and at 1.2% of orders it's too small to test reliably on this store's volume.

FINDING

Organic Search earns 40 to 43% of credit; Paid Search is undervalued by last click.

EVIDENCE

Seven models within 3 points; Markov interval 38.6% to 41.1%; 273,232 journeys.

INTERPRETATION

Big-channel credit is robust. Credit isn't incremental impact, and there's no spend data to judge return.

→ NEXT ACTION

Protect SEO; keep Paid Search until spend data allows a holdout (Part 3, test 4).



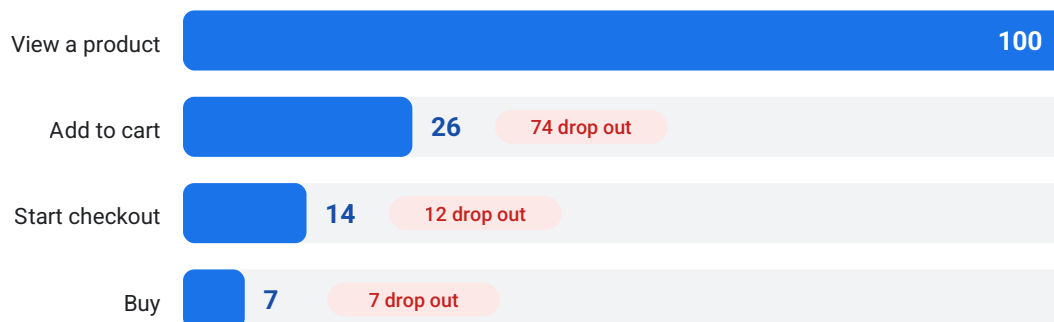
3. Where the journey loses buyers

74/100

product viewers leave before the basket. No clear device-level difference was detected.

Of 100 people who view a product, 74 leave before the basket

What happens to 100 product viewers · days with working cart tracking



Of 100 product viewers, 26 add to cart (25.7%), 14 start checkout and 7 buy. Observed, days with working cart tracking.

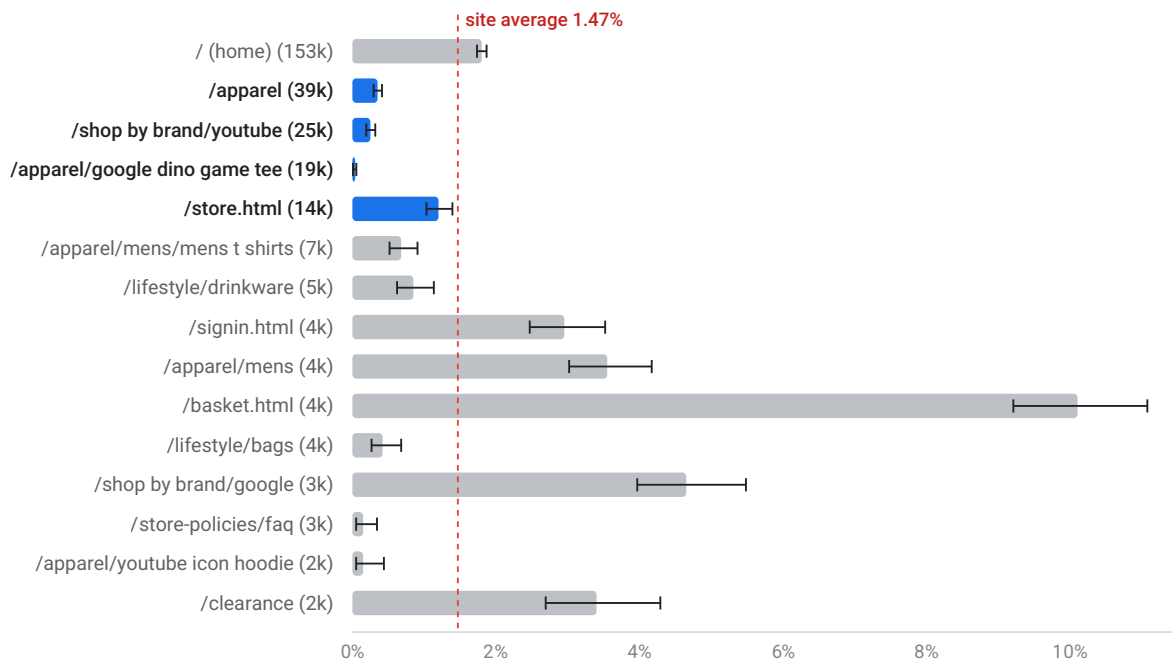
Supporting evidence: by device

STEP (SHARE OF THE PREVIOUS STEP)	DESKTOP	MOBILE	MOBILE - DESKTOP (95% INTERVAL)
Session to product view	20.8%	20.5%	-0.3 pts (-0.6 to 0.0)
Product view to cart	25.5%	26.1%	+0.6 pts (-0.1 to +1.3)
Cart to checkout	52.8%	53.3%	+0.5 pts (-1.1 to +2.1)
Checkout to purchase	47.2%	49.1%	+1.9 pts (-0.4 to +4.1)

Every interval includes zero. 1.35% of these sessions end in a purchase; the 1.47% conversion rate counts orders per session on all days.

Four big entry pages bring 29% of visits but 7% of purchases

Purchase rate with 95% Wilson interval, 15 most-used landing pages · blue = big and well below the site



Purchase rate by landing page; pages with 1,000+ sessions. Observed.

OBSERVED Four big entry pages (Apparel, the YouTube shop, the Dino Game Tee and store.html) bring 29% of visits and 7% of purchases; the Dino Game Tee had 18,653 visits and 4 orders.

POSSIBLE, NOT YET TESTED Low purchase intent (searches for Chrome's offline game), untagged links (63% of Apparel landings show as Direct), or friction on the page.

FINDING

3 in 4 product viewers never add to basket; big entry pages convert far below the site.

EVIDENCE

View to cart 25.7% on 59K product-view sessions; the four weak pages' intervals sit below 1.45%.

INTERPRETATION

The gap is real; its cause isn't established. Intent, tagging and design all fit.

NEXT ACTION

Check traffic mix, then A/B test the product page (Part 3, test 2); checkout after.

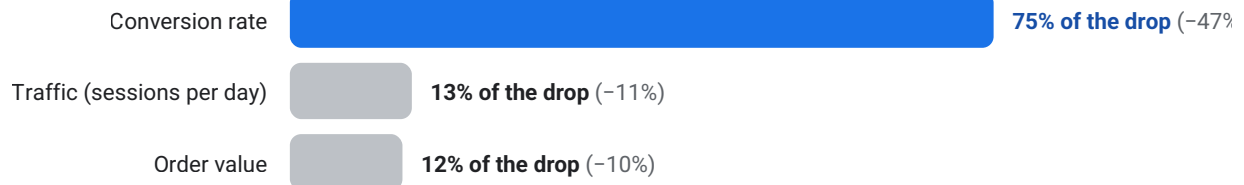
4. Why January fell

75%

of January's drop in revenue per day came from conversion, not traffic.

Conversion explains three quarters of January's 57% drop

Share of the December-to-January drop in revenue per day, split into traffic, conversion and order value



Revenue per day = sessions per day × conversion × order value, so the change splits exactly into three parts. Days with recorded order values. Observed.

	DECEMBER	JANUARY	CHANGE
Conversion rate	1.79%	0.98%	-0.81 pts
Sessions	133,351	118,377	-11%
Revenue per day*	\$5,101	\$2,174	-57%

*Days with recorded order values (26 to 31 January excluded). One holiday season: not proof that the pattern repeats every year.

FINDING

Visitors still came in January; far fewer bought.

EVIDENCE

On clean days: conversion -47% (75% of the drop), traffic -11% (13%), order value -10% (12%).

INTERPRETATION

A post-holiday drop in buying intent is the likeliest reading; three months can't rule out other causes.

NEXT ACTION

Set Q1 targets on a post-holiday baseline; track daily conversion against the forecast.



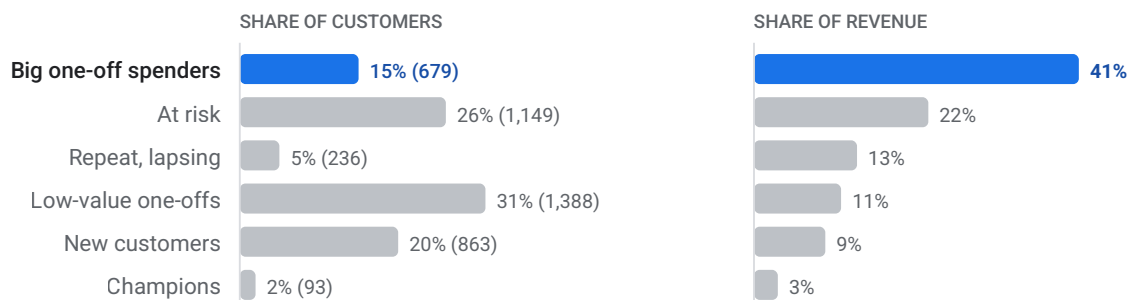
5. Customers and retention

41%

of revenue comes from 15% of customers, each from a single shopping trip. The store runs on one-off purchases.

15% of customers bring in 41% of revenue, each from a single shopping trip

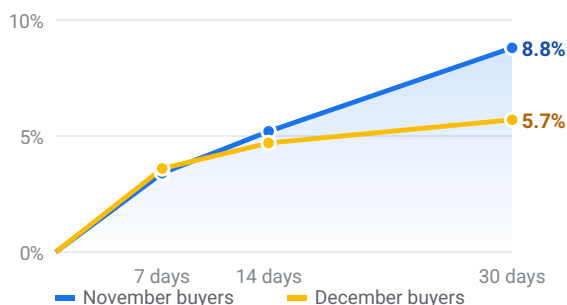
RFM segments: share of customers (with count) and share of revenue, 4,408 customers



RFM segments, 4,408 customers. Observed.

Holiday buyers fall behind after week one

Share of first-time buyers who buy again



FIRST PURCHASE	FOLLOWED	7 DAYS	14 DAYS	30 DAYS
November	1,532	3.4%	5.2%	8.8%
December	1,870	3.6%	4.7%	5.7%
January	too few	2.6%	3.6%	too few

Cells need 100+ customers followed for the full window.

7.5% of customers bought on more than one day; the median customer spent \$49; six in ten repeat buyers come back within two weeks. December buyers matched November's in week one (3.6% vs 3.4%), then fell behind, which fits gift buying. Browser-level IDs and three months both understate loyalty.

FINDING

One-off purchases dominate; holiday buyers rarely return.

EVIDENCE

7.5% repeat; 30-day repeat 8.8% (Nov) vs 5.7% (Dec), equal in week one.

INTERPRETATION

The window for a second order is short. Short history and browser IDs understate loyalty.

NEXT ACTION

Second-purchase offer within two weeks, against a holdout (Part 3, test 3).



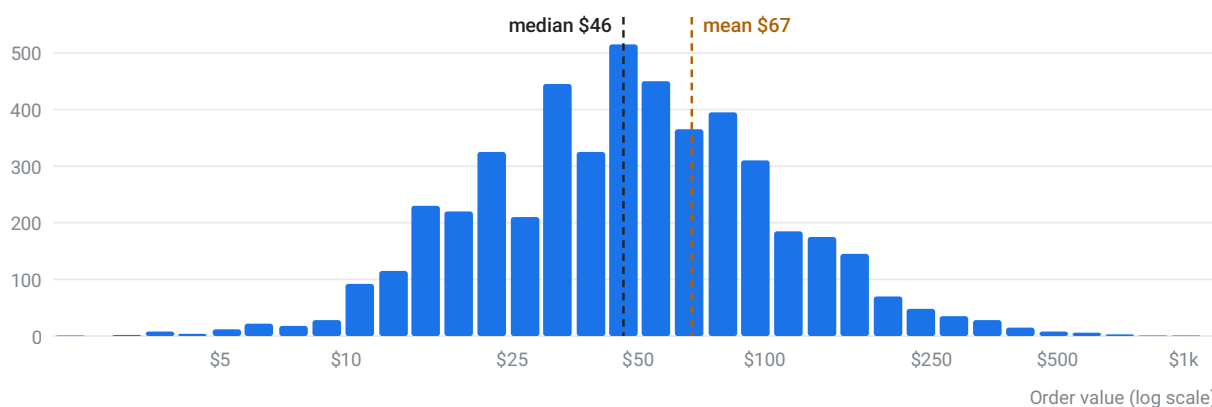
6. Orders and products

\$46

A typical order is \$46, and bigger baskets go with bigger orders.

A typical order is \$46; the top 10% of orders bring in 35% of revenue

Orders by value (log scale) · excludes 26 to 31 January, when order values weren't recorded



5,028 orders with recorded values: median \$46, mean \$67. Observed.

PRODUCTS BOUGHT TOGETHER IN 30+ ORDERS	ORDERS	LIFT
Light Pen Green + Light Pen Red	40	50×
Light Pen Blue + Light Pen Red	39	36×
Light Pen Blue + Light Pen Green	35	37×
Android Large Removable Sticker Sheet + Android SM S/F18 Sticker Sheet	30	29×
Crew Combed Cotton Sock + Crew Socks	30	12×

The top 10% of orders bring 35% of revenue, so revenue is broad. Orders with several products have a median of \$63 against \$25, and 59% of orders contain more than one product. Apparel is 47% of product revenue; 36 of 396 products make half of it. These are associations: they suggest what to test, not that bundles raise value.

FINDING

Basket size goes with order value; no cross-category pairs stand out.

EVIDENCE

Median \$46, mean \$67; multi-product median \$63 vs \$25; five pairs in 30+ orders.

INTERPRETATION

Sets of one item (pen colours) sell together; cross-category bundles have little support.

→ NEXT ACTION

Test a free-shipping threshold just above the median and multi-colour sets (Part 3, test 5).

PART 3

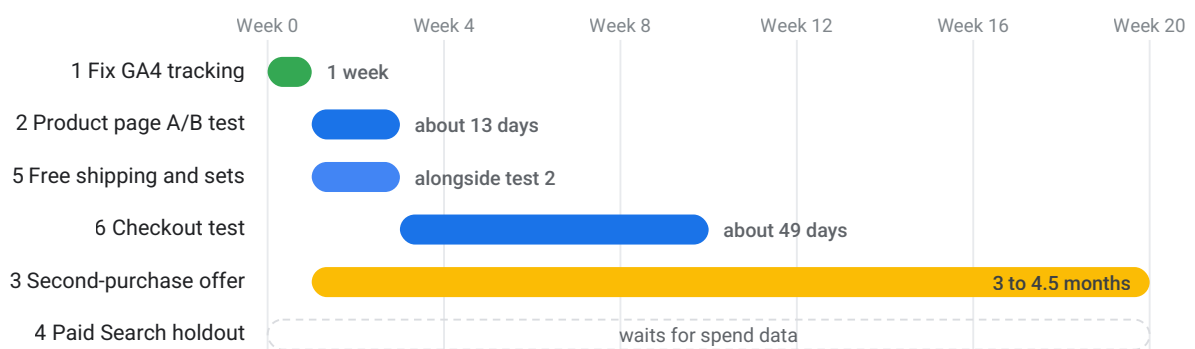
Recommendations and test plan

Each recommendation is a test with a success metric, a sample size and a run time, all worked out from the store's own January traffic (two arms, 95% confidence, 80% power). Detecting smaller lifts takes much longer, which shapes the order.

#	ACTION	SUCCESS METRIC	BASELINE	SIZE AND RUN TIME	EFFORT
1	Fix GA4: unwanted referrals, a transaction ID on every purchase, alerts when a tag stops firing	Self-referral share; share of purchases with an ID	30%; 84%	Verify within a week: near 0% and 100%	Low
2	A/B test the product page (delivery information, reviews, images), starting with the weak entry pages	Product view to cart	25.7%	10% lift: about 4,700 product-view sessions per arm, about 13 days. 5% lift: about 50 days	Medium
3	Second-purchase offer within two weeks of a first order	30-day repeat rate vs a holdout	5.7% to 8.8%	30% lift: 2,000 to 3,300 new customers per arm, about 3 to 4.5 months	Low
4	Protect SEO; keep Paid Search until a holdout can be run	Incremental orders in a geo holdout	64 last-click orders in 3 months	Too small to test on this volume; needs spend data and more traffic	Medium
5	Free-shipping threshold above \$46; multi-colour sets	Median order value	\$46	Size once the variance of the chosen threshold is known; run alongside test 2	Low
6	Checkout improvements, after the product page	Checkout to purchase	48.0%	Even a 10% lift needs about 1,700 checkouts per arm, about 49 days: test only big changes	Medium

Suggested sequence

Indicative run times from the sample sizes above; start dates assume each test begins once the tracking fix is verified



Run times from the power analysis on January daily volumes. Test 4 waits until spend data exists.

- i** The checkout test is slow because only about 70 checkouts happen a day in January, which is a second reason, besides the size of the leak, to start with the product page. Plan Q1 targets on a post-holiday baseline: the forecast in the appendix expects about 3,800 sessions a day.

Limits



Three months with the holidays in the middle: patterns may not repeat; customer value and forecasts have short histories.



GA4 identifies browsers, not people, so cross-device customers look less loyal and journeys look shorter.



Attribution splits credit; it doesn't prove cause. Only the tests above can measure impact.



About 9% of traffic sources are hidden by Google, and time of day is randomised in this sample.

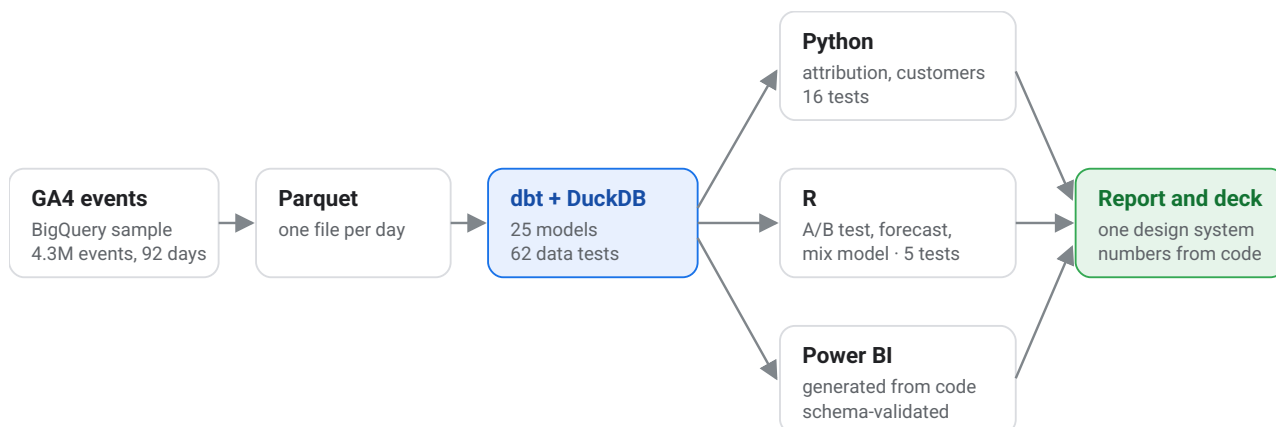
TECHNICAL APPENDIX



A. Data and pipeline

One command rebuilds everything

`python pipeline/run_all.py` · CI runs the unit tests and compiles dbt on every push



The data isn't in the repository; the first notebook downloads it from BigQuery.

Where each claim can be checked

CLAIM IN THIS REPORT	CODE	AUTOMATED CHECK
Duplicates add 7%; 837 orders without an ID recovered	<code>dbt int_ga4__purchases_deduped</code>	Unique order IDs; item revenue reconciles to orders

CLAIM IN THIS REPORT	CODE	AUTOMATED CHECK
Broken cart tag, missing order values	dbt int_ga4__tracking_quality, dim_date flags	Flags not null; exactly 22 broken-cart days and 6 missing-value days (26 to 31 Jan)
Self-referrals: 870 back to channels, 719 Unknown	dbt int_attribution__sessions, int_attribution__touchpoints	Every order hands out exactly one credit; models conserve conversions
Seven-model comparison; Markov shares and intervals	dbt mart_attribution_*; src attribution.py	6 unit tests incl. identical results for shuffled input
Device and landing-page intervals; basket lift; test sizes	src eda.py; notebook 04	6 unit tests (Wilson, differences, lift, sample size)
Segments, repeat timing, lifetime value	dbt dim_customers, customer_clv; src customers.py	Repeat timing matches buying days; 4 unit tests
Email test results	r/R/abtest.R, r/ab_test.R	5 R tests (sample ratio, differences, CUPED, power)
Dashboard numbers	powerbi/build_pbip.py (138 DAX measures)	Microsoft schema validation; every measure reference checked



B. Metric definitions and deduplication

METRIC	DEFINITION
Session	One GA4 session (user_pseudo_id + ga_session_id)
Order	One deduplicated purchase: by transaction ID, or by session, amount and basket when the ID is missing
Conversion rate	Orders ÷ sessions, all days
Funnel step rate	Sessions reaching a step ÷ sessions reaching the previous step, days with working cart tracking only
Purchase rate (landing pages)	Sessions with a purchase ÷ sessions
Customer	A browser (user_pseudo_id) with at least one order
Repeat within N days	Customers buying again within N days of their first buying day ÷ customers whose first purchase was at least N days before 31 January (cells need 100+ customers)
Revenue per day	Revenue ÷ days, days with recorded order values only
Channel	GA4 default channel grouping of the session; corrected attribution removes self-referrals, and journeys with no earlier visit are Unknown

Deduplication range

Only the 906 purchase events without a transaction ID are ambiguous



Every order with a transaction ID is deduplicated by that ID. Treating all 906 ID-less events as duplicates (the usual "dedupe by ID") gives 4,451 orders and \$307.6K; counting each as its own order gives 5,357 and \$339.5K. The rule used here (same session, amount and basket = one order) gives 5,288 and \$338.3K. Tested on orders that do have IDs, it merges about 3.6% of genuine repeat orders, so the true count is probably slightly above 5,288.



C. Methods

METHOD	USED FOR	KEY SETTINGS
Rule-based attribution (SQL)	Six models	30-day lookback; time decay half-life 7 days; position-based 40/20/40; last non-direct skips Direct and Unknown
Markov chain (Python)	Data-driven credit	273,232 journeys incl. non-converting; removal effects; 200 bootstrap resamples, fixed seed, sorted input
Wilson score interval	Single rates (landing pages)	95%
Normal approximation	Difference of two rates (devices)	95%
Log decomposition	January drop	Revenue per day = sessions per day × conversion × order value
RFM and BG/NBD	Segments, repeat likelihood	Fitted on Nov and Dec, checked on Jan (overshot: 114 expected vs 62 actual; top fifth found 39% of repeats)
Support, confidence, lift	Products bought together	Pairs in 30+ orders
Two-proportion power analysis	Test plan	Two-sided, α 0.05, power 0.8, January daily volumes
ARIMA with holiday flag	Traffic forecast	Chosen by two-week back-tests from five starts: 18.4% error on sessions, about 3,800 a day; daily orders not forecastable



D. Methods demonstrations on external datasets



The GA4 sample has no experiments and no ad spend, so two methods are shown on public datasets. Their numbers are not about the Google store.

External dataset



Email A/B test

Hillstrom (MineThatData, 2008): 64,000 customers of another retailer, randomised to a men's email, a women's email or none. Sample ratio check $p = 0.90$; all six results hold after a Holm correction. The men's email lifted purchases from 0.57% to 1.25% (+\$0.77 spend per customer over two weeks); the women's only moved past women's-wear buyers. CUPED removed at most 2.7% of variance; a 10% lift would need about 286,000 customers per group.

External dataset



Marketing mix model

Meta Robyn's simulated example data, 208 weeks. The model explains 85% of weekly revenue in unseen test weeks; outdoor is saturated (next dollar returns about \$0.37) while TV and print return about \$2.70 and \$1.90; reallocating the same budget within 70% to 150% of each channel could add about 11.5% media-driven revenue.

Charts and full method notes for both are in the repository (docs/experimentation.md, docs/mmm.md).



E. Dashboards and code

The Power BI report has five pages, each ending with the same finding, evidence, interpretation and next-action strip used in this report. Method notes for every section are in the repository's docs folder; every number here can be reprinted with `pipeline/headline_facts.py`.

[Overview](#)[Acquisition & attribution](#)[Conversion & landing pages](#)[Customers & retention](#)[Orders & products](#)

Data sources

Google Analytics 4 public sample e-commerce dataset (BigQuery public data); Kevin Hillstrom, MineThatData E-Mail Analytics and Data Mining Challenge (2008); Meta Robyn example data (simulated). Code under the MIT License. This is an independent analysis of Google's public GA4 sample and is not affiliated with or endorsed by Google.